

Evaluación y Análisis Matemático de Estrategias de Control para *Diaphorina citri* usando cultivo trampa a corto y largo plazo

Mujeres en Matemáticas
Creando vínculos de colaboración permanente

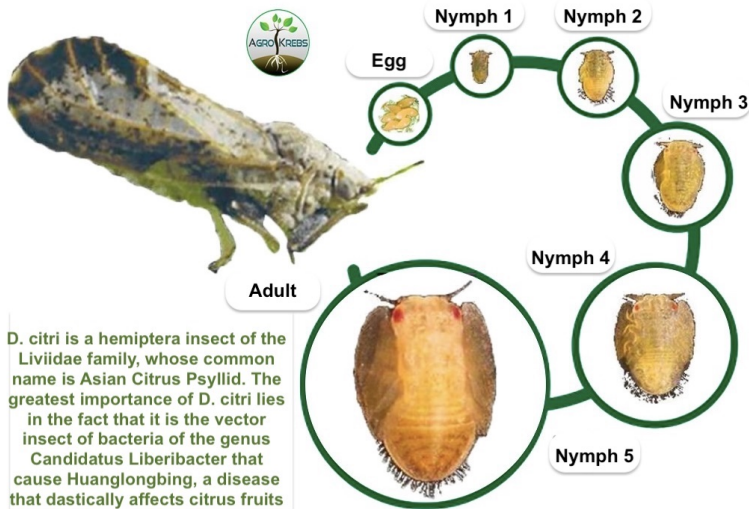
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3 de mayo de 2024



Diaphorina citri life cycle

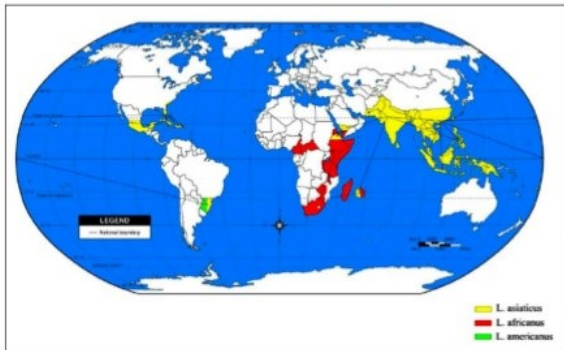


HLB worldwide in 2016



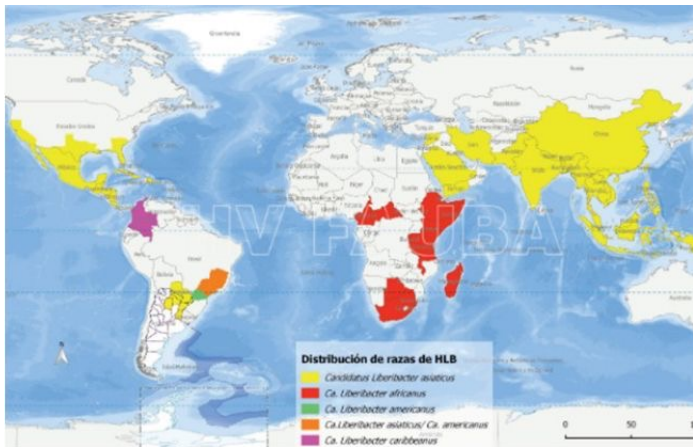
Distribución de las diferentes cepas de HLB

- Candidatus Liberibacter asiaticus* [Yellow box]
Candidatus Liberibacter africanus [Red box]
Candidatus Liberibacter americanus [Green box]



www.tratamientodeaguas.org.mx

HLB worldwide in 2022

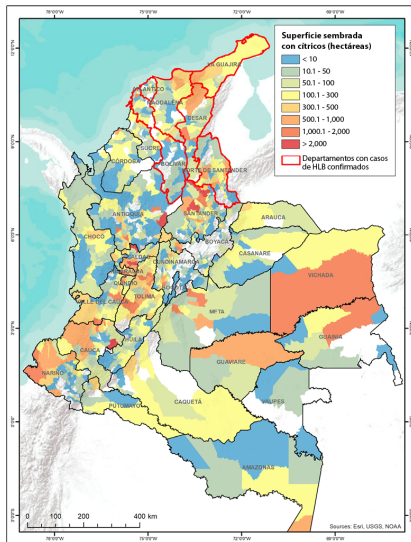


FUENTE: J.M. Bove, *Journal of pathology* (2006) y datos suministrados por el Programa Nacional de Prevención de Huanglongbing PNHuLB



Distribución mundial de enfermedad según razas del patógeno
Fuente SENASA Argentina

Diaphorina citri and HLB in Colombia



Diaphorina citri model in interaction with the shoots of a plant in a short term (naive model)

$$\begin{aligned}D_i'(t) &= f\phi D_m(B - \varphi_D D_i) - (\delta_i + \nu)D_i, \\D_m'(t) &= \nu D_i - \delta_m D_m, \\B'(t) &= \rho B \left(1 - \frac{B}{K}\right) - \varphi_i B D_i,\end{aligned}$$

where the initial conditions are

$$D_i(0) = D_{i_0}, D_m(0) = D_{m_0} \text{ y } B(0) = B_0,$$

and the variables are

- $D_i(t)$ is the average number of immature individuals, eggs and nymphs, of *Diaphorina citri* at time t ,
- $D_m(t)$ is the average number of individuals in the adult stage of *Diaphorina citri* at time t ,
- $B(t)$ is the average number of shoots on the host crop plant at time t ,

Diaphorina citri model parameters

Par	Description	Value
f	Proportion of female adult of <i>Diaphorina citri</i>	0.49
ϕ	Rate of immature per female adult of <i>Diaphorina citri</i> per shoot	4.09
$1/\varphi_D$	Average number of immature stage per shoot	17.15
ν	Transition rate from immature to adult state	0.05
δ_i	Immature state mortality rate	0.139
δ_m	Adult state mortality rate	0.0272
ρ	Intrinsic growth rate of plant shoots	0.1235
φ_i	Rate of affectation to the shoot by occupation and feeding of the immature state	0.0005
K	Shoot carrying capacity per plant	55

Region of biological interest

Theorem

$$\Omega_1 = \{(D_i, D_m, B) \in \mathbb{R}_+^3 : 0 \leq D_i \leq \frac{K}{\varphi_D}, 0 \leq D_m \leq \frac{\nu K}{\delta_m \varphi_D}, 0 \leq B \leq K\}$$

The region of biological interest, Ω_1 , is invariantly positive; that is, any trajectory that has an initial condition at Ω_1 will remain in Ω_1 for all $t \geq 0$.

Qualitative analysis

Equilibrium points

- 1 Extinction: $\mathcal{E}_1 = (0, 0, 0)$.
- 2 Extinction of *D. citri*: $\mathcal{E}_2 = (0, 0, K)$.
- 3 Coexistence of species: $\mathcal{E}_3 = (D_i^*, D_m^*, B^*)$:

$$D_i^* = \frac{\rho K(Q_0 - 1)}{Q_0(\varphi_i K + \rho \varphi_D)}, \quad D_m^* = \frac{\nu \rho K(Q_0 - 1)}{\delta_m Q_0(\varphi_i K + \rho \varphi_D)} \quad \text{and} \quad B^* = \frac{K(\varphi_i K + \rho \varphi_D Q_0)}{Q_0(\varphi_i K + \rho \varphi_D)},$$

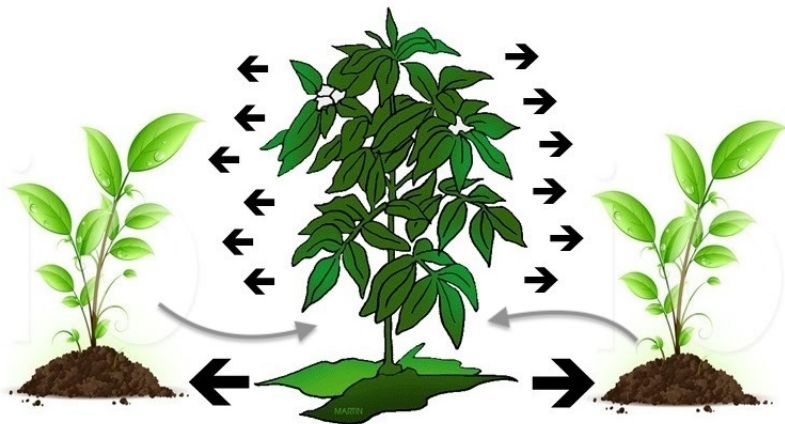
where

$$Q_0 = \frac{f\phi\nu K}{(\delta_i + \nu)\delta_m}.$$

Theorem

- 1 If $Q_0 < 1$, then $\mathcal{E}_1$ is unstable and $\mathcal{E}_2$ is asymptotically stable. ($\mathcal{E}_3$ makes no biological sense).
- 2 If $Q_0 > 1$, then $\mathcal{E}_1, \mathcal{E}_2$ are unstable and $\mathcal{E}_3$ is asymptotically stable.

Trap crop as a control



Population dynamics of *Diaphorina citri* in presence of two crops

$$D'_{ic}(t) = f\phi D_{mc}(B_c - \varphi_D D_{ic}) - (\delta_{ic} + \nu)D_{ic},$$

$$D'_{mc}(t) = \nu D_{ic} - (\delta_m + \gamma)D_{mc} + \lambda D_{ms},$$

$$B'_c(t) = \rho_c B_c \left(1 - \frac{B_c}{K_c}\right) - \varphi_i B_c D_{ic},$$

$$D'_{is}(t) = f\phi D_{ms}(B_s - \varphi_D D_{is}) - (\delta_{is} + \nu)D_{is},$$

$$D'_{ms}(t) = \nu D_{is} - (\delta_m + \lambda)D_{ms} + \gamma D_{mc},$$

$$B'_s(t) = \rho_s B_s \left(1 - \frac{B_s}{K_s}\right) - \varphi_i B_s D_{is},$$

with initial conditions

$$D_{ic}(0) = D_{ic0}, D_{mc}(0) = D_{mc0}, B_c(0) = B_{s0},$$

$$D_{is}(0) = D_{is0}, D_{ms}(0) = D_{ms0} \text{ and } B_s(0) = B_{s0}.$$

Region of biological interest

Theorem

$$\Omega_2 = \left\{ (D_{i_c}, D_{m_c}, B_c, D_{i_s}, D_{m_s}, B_s) : 0 \leq D_{i_x} \leq \frac{K_x}{\varphi_D}, 0 \leq D_{m_x} \leq A_x, 0 \leq B_x \leq K_x \right\}$$

where x takes the values of c or s depending on the type of crop and

$$A_s = \frac{\nu\gamma(K_s(\delta_m + \gamma) + K_c)}{\varphi_D((\delta_m + \lambda)(\delta_m + \gamma) - \lambda\gamma)},$$
$$A_c = \frac{\nu K_c + \lambda\varphi_D A_s}{\varphi_D(\delta_m + \gamma)}.$$

The region of biological interest, Ω_2 , is invariantly positive; that is, any trajectory that has an initial condition at Ω_2 will remain in Ω_2 for all $t \geq 0$.

Qualitative analysis

Equilibrium points

- 1 Extinction: $E_0 = (0, 0, 0, 0, 0, 0)$.
- 2 Extinction of *D. citri*:

$$E_1 = (0, 0, K_c, 0, 0, 0),$$

$$E_2 = (0, 0, 0, 0, 0, K_s),$$

$$E_3 = (0, 0, K_c, 0, 0, K_s).$$

- 3 Coexistence of species: $E_4 = (D_{ic}^*, D_{mc}^*, B_c^*, D_{is}^*, D_{ms}^*, B_s^*)$ which is the only biologically meaningful solution of this system of equations

$$K_c f \phi D_{mc} \left(1 - \left(\frac{\varphi_i}{\rho_c} + \frac{\varphi_D}{K_c} \right) \frac{(\delta_m + \gamma) D_{mc} - \lambda D_{ms}}{\nu} \right) - (\delta_{ic} + \nu) \frac{(\delta_m + \gamma) D_{mc} - \lambda D_{ms}}{\nu} = 0,$$

$$K_s f \phi D_{ms} \left(1 - \left(\frac{\varphi_i}{\rho_s} + \frac{\varphi_D}{K_s} \right) \frac{(\delta_m + \lambda) D_{ms} - \gamma D_{mc}}{\nu} \right) - (\delta_{is} + \nu) \frac{(\delta_m + \lambda) D_{ms} + \gamma D_{mc}}{\nu} = 0.$$

Stability

UNSTABLE equilibrium points

$E_0 = (0, 0, 0, 0, 0, 0), E_1 = (0, 0, K_c, 0, 0, 0),$ $E_2 = (0, 0, 0, 0, 0, K_s)$	Unstable
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Persistence of *D. citri*

$(0, 0, K_c, 0, 0, K_s)$	Unstable
$(D_{i_c}^*, D_{m_c}^*, B_c^*, D_{i_s}^*, D_{m_s}^*, B_s^*)$	Stable

Disappearance of *D. citri*

$(0, 0, K_c, 0, 0, K_s)$	Stable
Coexistence	Does not exist

Controls

Control	Description
w	Insecticide in commercial crop
u	Insecticide in trap crop
v	Pruning of shoots in trap crop

Optimal control problem

$$\min_{u,w,v} J(u, w, v) = \int_0^T \left(A_1(D_{m_c} + D_{i_c}) + A_2(D_{m_s} + D_{i_s}) + \frac{1}{2} (F_1 w^2 + F_2 u^2 + F_3 v^2) \right) dt$$

subject to

$$D'_{i_c}(t) = f\phi D_{m_c}(B_c - \varphi_D D_{i_c}) - (\delta_i + \epsilon_i w) D_{i_c} - \nu D_{i_c},$$

$$D'_{m_c}(t) = \nu D_{i_c} - (\delta_m + \epsilon_m w) D_{m_c} - (\gamma + \pi_c w + \kappa_c v) D_{m_c} + (\lambda + \pi_s u - \kappa_s v) D_{m_s},$$

$$B'_c(t) = \rho_c B_c \left(1 - \frac{B_c}{K_c} \right) - \varphi_i B_c D_{i_c},$$

$$D'_{i_s}(t) = f\phi D_{m_s}(B_{m_s} - \varphi_D D_{i_s}) - (\delta_i + \eta_i u) D_{i_s} - \nu D_{i_s},$$

$$D'_{m_s}(t) = \nu D_{i_s} - (\delta_m + \eta_m u) D_{m_s} - (\lambda + \pi_s u - \kappa_s v) D_{m_s} + (\gamma + \pi_c w + \kappa_c v) D_{m_c},$$

$$B'_s(t) = (\rho_s + \kappa_s v) B_s \left(1 - \frac{B_s}{K_s} \right) - \varphi_i B_s D_{i_s},$$

with initial conditions

$$D_{i_c}(0) = D_{i_c0}, D_{m_c}(0) = D_{m_c0}, B_c(0) = B_{s0}, D_{i_s}(0) = D_{i_s0}, D_{m_s}(0) = D_{m_s0} \text{ y } B_s(0) = B_{s0},$$

and

$$0 \leq u, w, v \leq 1.$$

Existence of controls

Theorem

There are admissible controls $(u^*, w^*, v^*) \in \mathcal{U}$ and the corresponding optimal states $(D_{i_c}^*, D_{m_c}^*, B_c^*, D_{i_s}^*, D_{m_s}^*, B_s^*)$ of the initial-value problem that minimizes the objective functional and satisfy that

$$J(u^*, w^*, v^*) = \min_{u, w, v} J(u, w, v).$$

Hamiltonian

$$\begin{aligned}\mathcal{H}(Z, u, \lambda) = & A_1(D_{m_c} + D_{i_c}) + A_2(D_{m_s} + D_{i_s}) + \frac{1}{2} (F_1 w^2 + F_2 u^2 + F_3 v^2) \\ & + \lambda_1 [f\phi D_{m_c}(B_c - \varphi_D D_{i_c}) - (\delta_i + \epsilon_i w) D_{i_c} - \nu D_{i_c}] \\ & + \lambda_2 [\nu D_{i_c} - (\delta_m + \epsilon_m w) D_{m_c} - (\gamma + \pi_c w + \kappa_c v) D_{m_c} + (\lambda + \pi_s u - \kappa_s v) D_{m_s}] \\ & + \lambda_3 \left[\rho_c B_c \left(1 - \frac{B_c}{K_c} \right) - \varphi_i B_c D_{i_c} \right] \\ & + \lambda_4 [f\phi D_{m_s}(B_s - \varphi_D D_{i_s}) - (\delta_i + \eta_i u) D_{i_s} - \nu D_{i_s}] \\ & + \lambda_5 [\nu D_{i_s} - (\delta_m + \eta_m u) D_{m_s} - (\lambda + \pi_s u - \kappa_s v) D_{m_s} + (\gamma + \pi_c w + \kappa_c v) D_{m_c}] \\ & + \lambda_6 \left[(\rho_s + \kappa v) B_s \left(1 - \frac{B_s}{K_s} \right) - \varphi_i B_s D_{i_s} \right]\end{aligned}$$

Controls characterization

$$\left\{ \begin{array}{l} w^*(t) = \min \left\{ 1, \max \left\{ \frac{\epsilon_i D_{ic}^* \lambda_1 + (\epsilon_m + \pi_c) D_{mc}^* \lambda_2 - \pi_s D_{ms}^* \lambda_5}{F_1}, 0 \right\} \right\} \\ u^*(t) = \min \left\{ 1, \max \left\{ \frac{\eta_i D_{is}^* \lambda_4 + (\eta_m - \pi_s) D_{ms}^* \lambda_5 + \pi_s D_{ms}^* \lambda_5}{F_2}, 0 \right\} \right\} \\ v^*(t) = \min \left\{ 1, \max \left\{ \frac{\kappa_c D_{mc}^* (\lambda_2 - \lambda_5) + \kappa_s D_{ms}^* (\lambda_2 - \lambda_5) - \kappa \lambda_6 B_s^* \left(1 - \frac{B_s^*}{K_s} \right)}{F_3}, 0 \right\} \right\} \end{array} \right\}$$

Estimated parameters

Dyn Par	Value
f	0.49
ϕ	4.09
$1/\varphi_D$	17.15
ν	0.05
δ_i	0.139
δ_m	0.0272
ρ_c	0.1235
ρ_s	0.2235
φ_i	0.0005
K_s	65
K_c	55
γ	0.6
λ	0.1

Con Effec	Value
ε_i	0.9
ε_m	0.85
π_c	0.2
π_s	0.3
η_i	0.9
η_m	0.9
κ	0.05
κ_c	0.05
κ_s	0.05

IC	Value
D_{ic0}	1
D_{mc0}	1
D_{is0}	1
D_{ms0}	1
B_{c0}	3
B_{s0}	3
A_1	20
A_2	1
F_1	500
F_2	100
F_3	50

Uncontrolled Dynamics

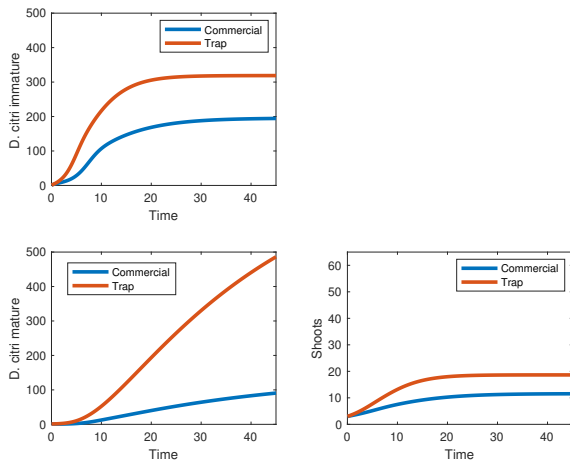


Figure: Dynamics of *Diphorina citri* in two crops, trap and commercial.

Cost effectiveness analysis

Relevant amounts

$$\bar{D}_{i_c} = \int_0^{t_f} D_{i_c}(t) dt = 6.522 \times 10^3,$$

$$\bar{D}_{m_c} = \int_0^{t_f} D_{m_c}(t) dt = 2.0039 \times 10^3,$$

$$\bar{D}_{i_s} = \int_0^{t_f} D_{i_s}(t) dt = 1.1617 \times 10^4,$$

$$\bar{D}_{m_s} = \int_0^{t_f} D_{m_s}(t) dt = 1.0204 \times 10^4.$$

Effectiveness

$$\begin{aligned} \text{Bene}_x = & \left(1 - \frac{1}{\bar{D}_{i_c}} \int_0^{t_f} D_{i_{c_x}}(t) dt \right) + \left(1 - \frac{1}{\bar{D}_{m_c}} \int_0^{t_f} D_{m_{c_x}}(t) dt \right) \\ & + \left(1 - \frac{1}{\bar{D}_{i_s}} \int_0^{t_f} D_{i_{s_x}}(t) dt \right) + \left(1 - \frac{1}{\bar{D}_{m_s}} \int_0^{t_f} D_{m_{s_x}}(t) dt \right) + \frac{1}{t_f K_c} \int_0^{t_f} B_{c_x}(t) dt. \end{aligned}$$

Results of the strategies

Stra.	$\int_0^{t_f} D_{i_c}(t)dt$	$\int_0^{t_f} D_{m_c}(t)dt$	$\int_0^{t_f} D_{i_s}(t)dt$	$\int_0^{t_f} D_{m_s}(t)dt$	$\int_0^{t_f} B_c(t)dt$
100	5.230×10^3	4.91×10^2	1.1583×10^4	5.809×10^3	5.75×10^2
010	6.436×10^3	9.68×10^2	9.830×10^3	7.90×10^2	4.31×10^2
001	6.468×10^3	1.310×10^3	1.3713×10^4	1.2387×10^4	4.30×10^2
110	4.277×10^3	2.30×10^2	8.515×10^3	4.48×10^2	6.87×10^2
101	4.951×10^3	3.77×10^2	1.3685×10^4	8.392×10^3	6.09×10^2
011	6.418×10^3	8.92×10^2	1.1621×10^4	8.88×10^2	4.33×10^2
111	4.305×10^3	2.31×10^2	1.0333×10^4	5.45×10^2	6.77×10^2
000	6.522×10^3	2.003×10^3	1.1617×10^4	1.0204×10^4	4.22×10^2

Strategy with the best effectiveness (110)

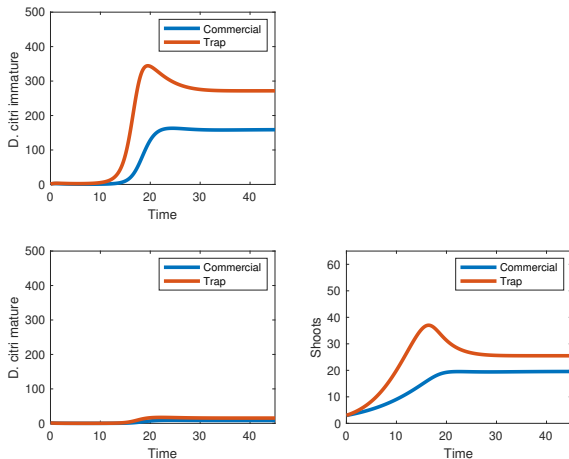


Figure: Insecticide in both crops without pruning.

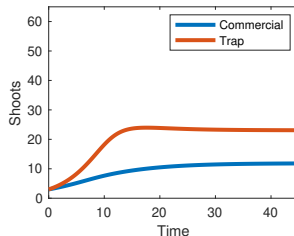
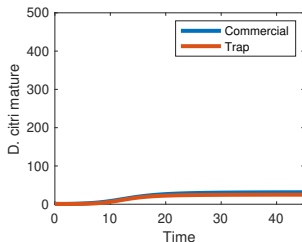
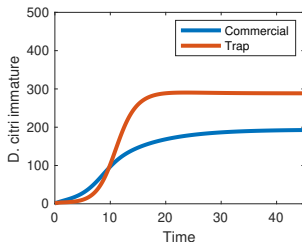
Best cost-effectiveness strategy

$$\text{ACER} = \frac{\text{Strategy cost}}{\text{Strategy effectiveness}}$$

Results

Strat	Effectiveness	Cost	ACER
100	1,61890	22500	1.3897×10^5
010	1,7803	4500	2.527×10^4
001	0,1337	2250	1.6833×10^5
110	2,7302	27000	9.889×10^4
101	1,2982	24750	1.9064×10^5
011	1,6581	6750	4.071×10^4
111	2,5555	29250	1.1445×10^5

The best strategy according to ACER (010)



Conclusions

We propose a model that represents the population dynamics of *D. citri* in interaction with the shoots of two crops, trap and commercial. In addition, this system is controlled by the use of insecticides in both crops and pruning in the trap crop.

If controls are not exercised on the *D. citri* population, it will establish itself over time with high levels of individuals per shoot, and the shoots will not reach their carrying capacity.

Applying controls to their maximum capacity, the best cost-effective strategy is to use insecticide only on the trap crop.

Diaphorina citri model in interaction with the shoots of a plant in a long-term

$$\begin{aligned}D'_i(t) &= f\phi D_m(B - \varphi_D D_i) - (\delta_i + \nu)D_i, \\D'_m(t) &= \nu D_i - \delta_m D_m, \\B'(t) &= \alpha BS - \mu B - \varphi_i B D_i, \\S'(t) &= \rho S \left(1 - \frac{S}{K}\right) - \beta BS,\end{aligned}$$

where initial conditions are $D_i(0) = D_{i_0}$, $D_m(0) = D_{m_0}$, $B(0) = B_0$ y $S_0(0) = S_0$ and the variables are

- $D_i(t)$ is the average number of immature individuals, eggs and nymphs, of *Diaphorina citri* at time t ,
- $D_m(t)$ is the average number of individuals in the adult stage of *Diaphorina citri* at time t ,
- $B(t)$ is the average number of shoots on the host crop plant at time t ,
- $S(t)$ is the chemical stimulus in the plant responsible for the sprouting process.

Parameters of the long-term dynamics of a plant's shoots

Par	Description	Value
ρ	Intrinsic growth rate of plant shoots	0.1829
α	growth rate of the shoots due to the stimulus	0.1472
β	rate of decrease in stimulus due to sprouting	0.0903
μ	transition rate from shoot to branch	0.0914
φ_i	Rate of affectation to the shoot by occupation and feeding of the immature state	5×10^{-6}
K	Shoot carrying capacity per plant	20

Region of biological interest

Let

$$\Omega_3 = \left\{ (D_i, D_m, B, S) \in \mathbb{R}^4 : 0 \leq D_i \leq R, \quad 0 \leq D_m \leq \frac{vR}{\delta_m}, \quad 0 \leq B \leq \varphi_D R, \quad 0 \leq S \leq K \right\}$$

where $Q_B = \frac{\alpha K}{\mu}$ and $R = \frac{\mu(Q_B - 1)}{\varphi_i}$.

Theorem

The region of biological interest, Ω_3 , is invariantly positive; that is, any trajectory that has an initial condition at Ω_3 will remain in Ω_3 for all $t \geq 0$.

Qualitative analysis

Equilibrium points

- 1 Extinction: $\mathcal{E}_1 = (0, 0, 0, 0)$.
- 2 No sprouting: $\mathcal{E}_2 = (0, 0, 0, K)$.
- 3 Extinction *D. citri*: $\mathcal{E}_3 = (0, 0, B^*, S^*)$:
where $B^* = \left(1 - \frac{1}{Q_B}\right) \frac{\rho}{\beta}$ and $S^* = \frac{\mu}{\alpha}$.
- 4 Coexistence of species: $\mathcal{E}_4 = (\overline{D}_i, \overline{D}_m, \overline{B}, \overline{S})$:

$$\overline{D}_i = \frac{\rho\mu Q_B(Q_O - 1)}{Q_D(\alpha K\beta\varphi_D + \rho\varphi_i)}, \quad \overline{D}_m = \frac{\rho\mu\nu Q_B(Q_O - 1)}{\delta_m Q_D(\alpha K\beta\varphi_D + \rho\varphi_i)},$$
$$\overline{B} = \frac{\rho}{Q_D} \left(\frac{1}{\beta} + \frac{\varphi_D\mu Q_B(Q_O - 1)}{\alpha K\beta\varphi_D + \rho\varphi_i} \right) \text{ and } \overline{S} = \frac{K}{Q_D} \left(Q_D - 1 - \frac{\varphi_D\beta\mu Q_B(Q_O - 1)}{\alpha K\beta\varphi_D + \rho\varphi_i} \right),$$

where

$$Q_D = \frac{f\phi\rho\nu}{\beta(\delta_i + \nu)\delta_m} \quad \text{and} \quad Q_O = Q_D \left(1 - \frac{1}{Q_B} \right).$$

Qualitative analysis

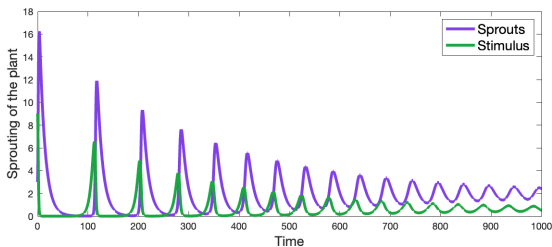
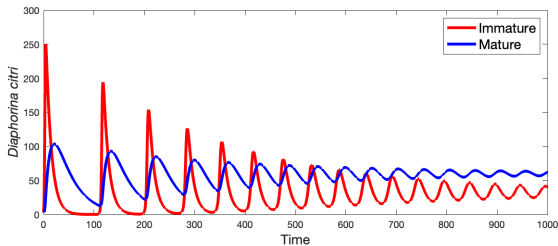
Theorem

Equi. Point.	$Q_B < 1$	$1 < Q_B < \lambda$	$\lambda < Q_B$
	$Q_O < 0$	$0 < Q_O < 1$	$Q_O > 1$
ε_1	Unstable	Unstable	Unstable
ε_2	Asym. stable	Unstable	Unstable
ε_3	X	Asym. stable	Unstable
ε_4	X	X	Asym. stable

where

$$\lambda = \left(1 - \frac{1}{Q_D}\right)^{-1}$$

Dynamics of *D. citri* in the long term



Acknowledgments

